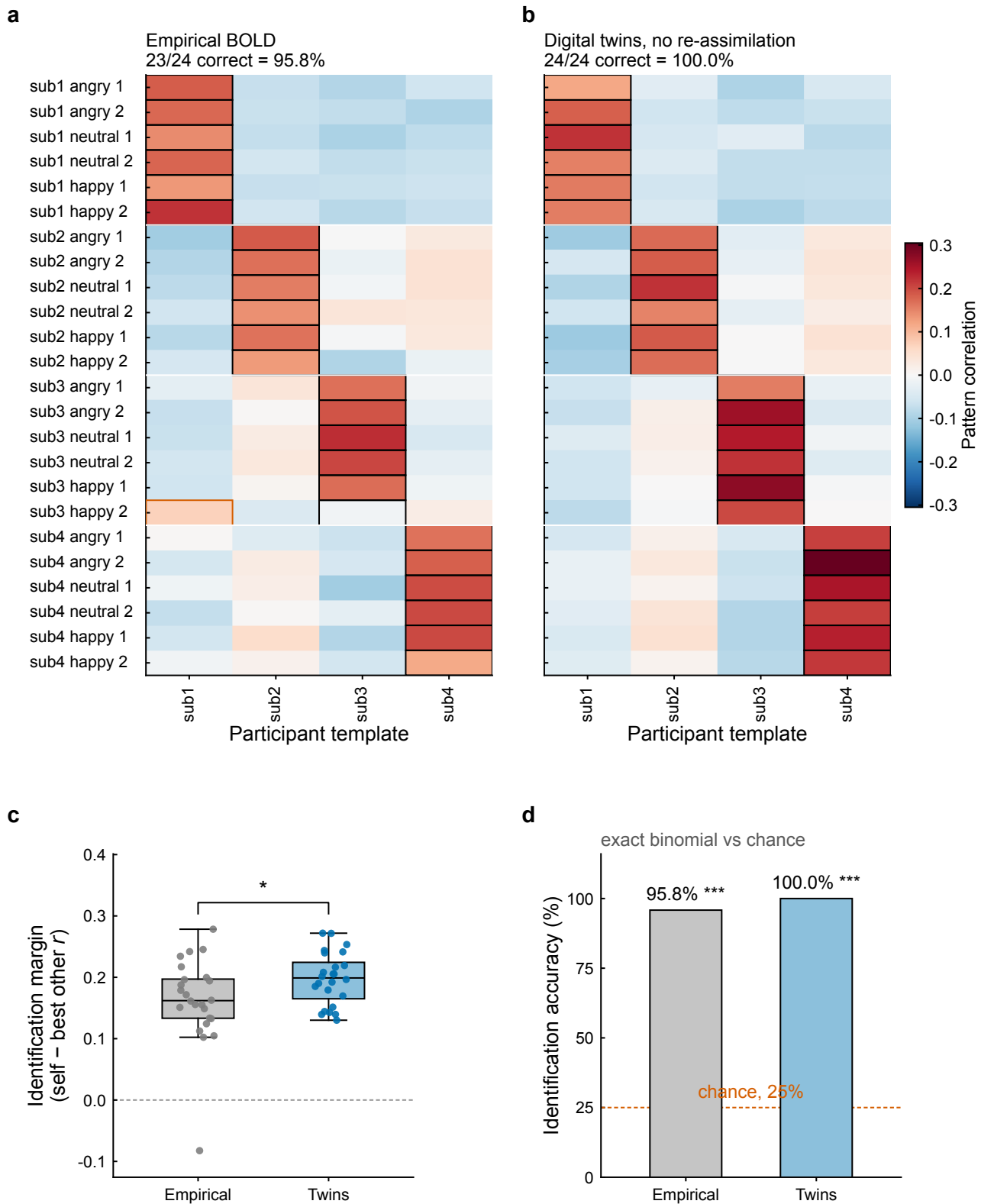




Supplementary Fig. S16 | Digital twin brains preserve participant-specific signatures on held-out trials without re-assimilation. Four participants (sub1 and sub2, healthy controls; sub3, alcohol use disorder; sub4, major depressive disorder) x 6 held-out emotional-face trials ($n = 24$ trials; unit of observation, one held-out trial; the labels sub1-sub4 are local to this figure). Hyperparameters were estimated from the first half of the task and the second half was simulated forward without re-assimilation, so every panel tests generalisation rather than fit. **a**, Pearson correlation between each held-out trial's activation pattern and each participant's template, from empirical BOLD; colour scale symmetric at ± 0.30 , black outline marks the correct template and an orange outline the template actually selected when identification failed; no error indicator and no inferential test (descriptive similarity matrix). **b**, The same matrix from the digital twins. **c**, Identification margin per trial (self minus best-other correlation); boxes show the median and interquartile range with whiskers at $1.5 \times$ IQR and every trial plotted; empirical 0.1628 ± 0.0699 versus twins 0.1974 ± 0.0426 , two-sided paired t test, $t(23) = 2.293$, $*P = 0.03134$ (the asterisk in the panel marks this test; Wilcoxon signed-rank $P = 0.01787$), no multiple-comparison correction (one pre-specified contrast). Within each data type the self correlation exceeds the best other correlation: empirical 0.1662 versus 0.0034 , $t(23) = 11.411$, $P = 5.99 \times 10^{-11}$; twins 0.2021 versus 0.0047 , $t(23) = 22.725$, $P = 2.94 \times 10^{-17}$. **d**, Identification accuracy against the 25% chance level of a four-way choice; bars show the percentage of trials identified correctly, with no error bar because the quantity is a count over all 24 trials: empirical 23/24 = 95.8% (the single failure is a happy trial of sub3, assigned to the sub1 template) and twins 24/24 = 100.0%; one-sided exact binomial test against 0.25, $P = 2.59 \times 10^{-13}$ and $P = 3.55 \times 10^{-15}$.